

Alvi Ataur Khalil

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CURRENT POSITION

Southern Illinois University, Carbondale, IL, USA

Aug. 2025 - Present

Tenure-Track Assistant Professor, Computer Science, School of Computing

KEY ACCOLADE/ACCOMPLISHMENTS

- Published over twenty peer-reviewed papers (16 as first author) in top-tier venues including ACSAC, DSN, ISQED, IEEE CNS, IEEE LCN, IEEE ICMLA, IEEE SMC, ACM DLT, and Elsevier Computer Networks, spanning blockchain security, UAV autonomy and defense, and the privacy and security of large language models.
- Principal Investigator on the NSF NAIRR Pilot Award (#NAIRR260054), securing 10,000 GPU-hours on NCSA's Delta supercomputer, the NSF I-Corps Award (#2637735), securing \$50K USD for deep tech commercialization, and the IEEE Computer Society Emerging Technology Grant (#2026-CS-AGORA), securing \$25K USD for a hands-on community lab on the security of autonomous AI agents.
- Awarded one U.S. Patent (US 12,566,450 B2) on UAV trajectory planning against hostile defense systems.
- Selected as an IFLIP2 Fellow under the NSF IUSE initiative (Award #2439762) for contributions to STEM education reform initiative and received multiple awards, including Outstanding Graduate Scholar of the Year at FIU (2023-24), Second Place in Engineering at the FIU Graduate School Scholarly Forum 2023, IEEE Upsilon Pi Epsilon (UPE) Honor Society Scholarship 2024, and the NSF Student Travel Grant.

RESEARCH INTERESTS

- Trustworthy and secure AI, focusing on vulnerabilities of large language models (LLMs), vision-language models (VLMs), and retrieval-augmented generation (RAG) systems, such as knowledge-corpus poisoning and adversarial attacks, along with robust defenses and reliability evaluation of AI systems.
- Privacy-preserving machine learning, including federated and split learning for multimodal and LLM-based models in sensitive domains such as healthcare.
- Blockchain security, with a focus on vulnerabilities in decentralized off-chain Layer 2 (L2) systems, such as optimistic rollups and payment channel networks, and their mitigation.
- AI-driven autonomous systems, particularly reinforcement learning (RL) for secure UAV navigation, and data-driven threat detection for cyber-physical systems (CPSs).

FUNDED RESEARCH PROJECTS

I-Corps: Translation potential of Automated AI Reliability Evaluation Platform

Sep. 2026 – Aug. 2027

- Funding organization: National Science Foundation (NSF)
- Role: Principal Investigator (PI)
- Award ID: 2637735
- Award Amount: \$50,000 USD to accelerate the commercialization of our deep-tech innovation.

AGORA: An Agentic-AI Open Red-team & Assurance Community Lab

Aug. 2026 – Dec. 2027

- Funding organization: IEEE Computer Society (IEEE CS) Emerging Technology Grants Program 2026
- Role: Principal Investigator (PI)
- Award Amount: \$25,000 USD to run a hands-on community lab on the security of autonomous AI agents.

SplitMind: Multimodal AI via Split Learning for Privacy-Critical Visual Intelligence

May 2026 – May 2027

- Funding organization: National Science Foundation (NSF)'s National Artificial Intelligence Research Resource (NAIRR) Pilot Resource Grant
- Role: Principal Investigator (PI)
- Award ID: NAIRR260054
- Resources: 10,000 GPU Hours on NCSA Delta GPU (Delta GPU)

Florida International University (FIU) , Miami, FL, USA M.S. leading to Ph.D. in Electrical and Computer Engineering	Aug. 2020 - Aug. 2025
Florida International University (FIU) , Miami, FL, USA M.S. in Computer Engineering	Aug. 2020 - Jul. 2024
Khulna University of Engineering and Technology (KUET) , Khulna, Bangladesh B.Sc. in Computer Science and Engineering	Apr. 2014 - Mar. 2018

PUBLICATIONS

US Patent

- [1] M. Rahman and **A. Khalil**. “Federated Deep Reinforcement Learning-Assisted UAV Trajectory Planning Against Hostile Defense System.” Patent No. **US 12,566,450B2**; Published: Jul 10, 2025; Granted: **Mar 3, 2026**.

Journal Papers

- [2] T. Rajabi*, **A. Khalil***, M. Manshaei, M. Rahman, M. Dakhilalian, M. Ngouen, M. Jadliwala, and A. S. Uluagac. “Feasibility Analysis for Sybil Attacks in Shard-Based Permissionless Blockchains.” *ACM Distributed Ledger Technologies: Research and Practice (DLT)*, Dec 2023.
- [3] **A. Khalil**, M. Selim, and M. Rahman. “Deep Learning-based Energy Harvesting with Intelligent Deployment of RIS-assisted UAV-CFmMIMOs.” *Elsevier Computer Networks*, Jun 2023. (IF: 4.4)
- [4] A. Jakaria, M. Rahman, M. Asif, **A. Khalil**, H. Kholidy, M. Anderson, and S. Drager. “Trajectory Synthesis for a UAV Swarm Based on Resilient Data Collection Objectives.” *IEEE Transactions on Network and Service Management (TNSM)*, Oct 2022. (IF: 4.7)

Conference Papers (Blockchain Security)

- [5] **A. Khalil** and M. Rahman. “xPoz-Hub: Privacy Infiltration in Payment Channel Hubs through Balance Probing and Transaction Discovery.” *ACSAC*, 2025.
- [6] **A. Khalil**, M. Hasan, and M. Rahman. “OptiGame: Game-Theoretic Enhancements for Optimistic Rollups through Bayesian Strategies.” *EAI SecureComm*, 2025.
- [7] **A. Khalil** and M. Rahman. “RollGuard: Defending RPC Manipulation Attacks in Optimistic Rollups with Graph-Based AI.” *26th International Symposium on Quality Electronic Design (ISQED)*, 2025.
- [8] **A. Khalil** and M. Rahman. “PAROLE: Profitable Arbitrage in Optimistic Rollup with ERC-721 Token Transactions.” *54th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, Aug 2024. (Acceptance rate: 20%)
- [9] **A. Khalil**, M. Rahman, and H. Kholidy. “FAKey: Fake Hashed Key Attack on Payment Channel Networks.” *IEEE Conference on Communications and Network Security (CNS)*, Oct 2023. (Acceptance rate: 28%)
- [10] **A. Khalil**, J. Franco, I. Parvez, A. S. Uluagac, H. Shahriar, and M. Rahman. “A Literature Review on Blockchain-enabled Security and Operation of Cyber-Physical Systems.” *IEEE 46th Annual Computers, Software, and Applications Conference (COMPSAC)*, Aug 2022.

Conference Papers (Privacy and Security of Large Language Models)

- [11] A. Kafi and **A. Khalil**. “RAG-Narok: Retrieval-Aware Knowledge Corpus Poisoning in RAG with Source-specific Refutation.” *IEEE ICMLA*, 2026. [Accepted]
- [12] M. Syfullah and **A. Khalil**. “U-Shaped Split Learning for Visual Question Answering with Contribution-Aware Weighted Aggregation.” *IEEE SMC*, 2026. [Accepted]
- [13] M. Syfullah and **A. Khalil**. “LLM-Based Schema-Aware Split Learning for Privacy-Preserving Mental Distress Prediction Across Heterogeneous Surveys.” *IEEE HealthCom*, 2026. [Accepted]

Conference Papers (UAV: Autonomous Navigation, Energy Harvesting, and Security)

- [14] A. Alva, L. Martinez Moreno, M. Asif, **A. Khalil**, M. Rahman, A. Cuzzocrea, and S. Hossain. “Secured UAV Navigation: A Novel Intrusion Detection System Based on PWM Signal Analysis.” *11th IEEE Swiss Conference on Data Science (SDS)*, Sep 2024.

- [15] A. Khalil and M. Rahman. “Adaptive Neuro-Fuzzy Inference System-Based Lightweight Intrusion Detection System for UAVs.” *IEEE 48th Conference on Local Computer Networks (LCN)*, Sep 2023. (Acceptance rate: 26%)
- [16] A. Khalil and M. Rahman. “FeD-UP: Federated Deep Reinforcement Learning-based UAV Path Planning against Hostile Defense System.” *18th International Conference on Network and Service Management (CNSM)*, Dec 2022. (Acceptance rate: 19%)
- [17] A. Khalil*, A. Byrne*, M. Rahman, and M. Manshaei. “REPlanner: Efficient UAV Trajectory-Planning using Economic Reinforcement Learning.” *IEEE International Conference on Smart Computing (SMARTCOMP)*, Oct 2021. (Acceptance rate: 30%)
- [18] A. Khalil, M. Selim, and M. Rahman. “CURE: Enabling RF Energy Harvesting using Cell-Free Massive MIMO UAVs Assisted by RIS.” *IEEE Conference on Local Computer Networks (LCN)*, Sep 2021. (Acceptance rate: 28%)

Other Conference Papers

- [19] S. Syed, A. Khalil, K. D. Gupta, S. Jabeen, and M. Rahman. “Query-Based Model Extraction Attack on GCN: A Surrogate Model Technique for Non-Euclidean Data.” *AAAI Fall Symposium Series (FSS)*, 2025.
- [20] M. Shahriar, A. Khalil, M. Rahman, M. Manshaei, and D. Chen. “iAttackGen: Generative Synthesis of False Data Injection Attacks in Cyber-physical Systems.” *IEEE Conference on Communications and Network Security (CNS)*, Feb 2022. (Acceptance rate: 26%)
- [21] A. Khalil, A. Mahmud, and A. Ahmed. “FicBit: An Improved Fractal Image Compression using Bio-Inspired Techniques.” *5th International Conference on Electrical Information and Communication Technology (EICT)*, Mar 2022.
- [22] N. Haque*, A. Khalil*, M. Rahman, M. Amini, and S. Ahamed. “BIOCAD: Bio-Inspired Optimization for Classification and Anomaly Detection in Digital Healthcare Systems.” *IEEE International Conference on Digital Health (ICDH)*, Sep 2021. (Acceptance rate: 20%)

*Co-first author

POSTERS/ABSTRACTS

- Represented FIU at the Academy of Science, Engineering, and Medicine of Florida (ASEMFL) by presenting poster titled “State-root manipulation attack in optimistic rollup through updating the remote procedure call (RPC) address” at the Annual Meeting 2024.
- Presented a poster titled “SHiP: Securing Hashed Timelock Contracts in Payment Channel Networks” at the 2023 IEEE Conference on Communications and Network Security (CNS 2023).
- Presented a poster titled “Data-driven ANFIS-based IDS for UAV” at the FIU UGS Scholarly Forum 2023.

HONORS AND AWARDS

- Awarded the **Outstanding Graduate Scholar of the Year** by Outstanding Student Life Awards organized by the Division of Academic & Student Affairs at FIU for research contributions in the academic year 2023-24.
- Awarded FIU University Graduate School (UGS) **Dissertation Year Fellowship** for Spring and Summer 2025.
- Awarded **NSF Student Travel Grant** for the 2023 IEEE CNS, acknowledging academic excellence and research potential in secured communications and networking (Award #2331646).
- Awarded the Upsilon Pi Epsilon (UPE) Honor Society Scholarship 2024 from the Computing Sciences and the IEEE Computer Society (the largest association of computer professionals).
- Honored with **Second Place (among 81 posters)** in the Engineering category for the poster titled “Data-driven ANFIS-based IDS for UAV” at the FIU GSAW Scholarly Forum 2023.
- Completed the Fall 2021 **NSF I-Corps cohort as Entrepreneurial Lead** of team 2426, funded by National Science Foundation (NSF) under award #2138301. (Credential).
- Inducted by the Honor Society of **Phi Kappa Phi** by the election of the chapter at FIU for outstanding academic achievements in the academic year 2020-2021.
- Deans List award of the Faculty of EEE for Excellent Academic Performance in the 2016-17 and 2015-16 sessions.

WORK EXPERIENCE

Analytics for Cyber Defense (ACyD) Laboratory, FIU, Miami, FL, USA

Aug. 2020 - Jul. 2025

Graduate Assistant, Department of Electrical and Computer Engineering

- Collaborated and made substantive contributions to 1 NSF project, 1 NSA project, and 1 DOE project.
- Articulated defense mechanisms against cyber attacks on off-chain L2 solutions for **Blockchain**.
- Designed intelligent navigation for military drones/UAVs in hostile territory using reinforcement learning.
- Articulated deep learning-based energy harvesting models for IoT devices using cell-free massive MIMOs, and intelligent reflecting surfaces.
- Mentored 7 undergraduate and graduate researchers supported by NSF/NSA/DOE projects.

MazeGeek Technologies BD Ltd., Dhaka, Bangladesh

Aug 2018 - Jul. 2019

Software Developer

- Developed and launched 5 Android applications that collectively achieved over 1,000 downloads, enhancing user engagement by 15% within the first six months of release.
- Designed robust project architectures that streamlined application functionality, resulting in a 10% reduction in development time and improved performance.
- Managed client requirements and documentation, ensuring 100% adherence to project specifications and timelines.
- Participated in key client meetings, fostering relationships and securing 2 new contracts through effective communication and project presentations.

Daffodil International University (DIU), Dhaka, Bangladesh

May 2018 - Jun. 2018

Lecturer, Department of Computer Science and Engineering

- Conducted classes and lab sessions for over 60 students in Artificial Intelligence (CSE 412 & 413), achieving a 90% student satisfaction rating through engaging instructional methods.
- Implemented hands-on projects that improved student proficiency in GUI development, resulting in 80% of students completing practical assignments on time.

PROFESSIONAL SERVICES

External Services

Conference Chair/Committee Member

- Served as the Track Chair for Track 4: Network and Communication Security and Privacy of the IEEE SaTC 2027 Conference, focusing on IoT network and communication security, blockchain for IoT, secure IoT protocols, and next-generation IoT security.
- Lead the panel “Trustworthy AI in Practice: Emerging Threats and Privacy-Preserving Solutions for Illinois Healthcare and High-Stakes Systems” at the 2026 Illinois Innovation Network (IIN) Conference, 2026.
- Served as a member of the Technical Program Committee of the **NeurIPS 2026 Workshop** on “Secure and Trustworthy Quantum Machine Learning” (SaTQuML).
- Served as a member of the Technical Program Committee of the 38th IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2026).
- Served as a member of the Technical Program Committee of the Workshop on “AI-Native Networking–Compute Co-Design for Emerging 6G Applications” at the IEEE Consumer Communications and Networking Conference (CCNC 2027).
- Served as a member of the Technical Program Committee of the 23rd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (IEEE MASS 2026).
- Served as the **Co-Chair** of the Special Session on “Graph Machine Learning for Complex Systems and Applications” at the International Conference on Machine Learning and Applications (ICMLA 2026).
- Served as the **Co-Chair** of the Special Session on “Advances in Multimodal AI: Method and Application” at the IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC 2026).
- Served as the **Co-Chair** of the Special Session on “Multi-modal Machine Learning in Practice” at the International Conference on Machine Learning and Applications (ICMLA 2026).
- Served as a member of the Scientific Program Committee for the 12th IEEE Swiss Conference on Data Science (SDS 2025), taking place on June 26-27, 2025, in Zürich, Switzerland.

- Served as a member of the Technical Program Committee for the Special Session on Multi-modal Machine Learning (MAPLE) at International Conference on Machine Learning and Applications (ICMLA 2025).

Invited Talks

- Delivered an invited seminar talk on **Cybersecurity and Blockchain** for the Department of Computer Science Seminar Series at Missouri State University, November 2025.

University Service

Southern Illinois University Carbondale

- Member, Faculty Senate Undergraduate Education Policy Committee (Fall 2026 - Present).
- Member, Faculty Senate Elections Committee (Fall 2026 - Present).
- Faculty Volunteer, Class of 2030 Freshman Convocation (Fall 2026).
- AI Undergraduate Studies Committee [Curriculum] (Fall 2026 - Spring 2027).
- CS Undergraduate Studies Committee [Curriculum / Assessment / Accreditation] (Fall 2025 - Spring 2027).

TEACHING EXPERIENCE

Assistant Professor , Southern Illinois University Carbondale	Aug. 2025 - Present
• Digital Forensics with Graduate and Undergraduate Sections (CS 413/513) • Problem Solving with Computers (CS 201)	
Instructor , Florida International University	May. 2024 - Dec. 2024
• Network Protocols for Internet of Things (CNT 4165)	
Graduate Teaching Assistant , Florida International University	Aug. 2020 - Apr. 2024
• Senior Design Projects (EEL 4920) - Served as a mentor for over 100 capstone projects, challenging students to apply their academic knowledge to real-world projects and guiding through project planning, design, and implementation phases. • Digital Forensics (EEL 4802)	
Lecturer , Daffodil International University	May 2018 - Jun. 2018
• Artificial Intelligence (CSE 412 & 413)	

PROFESSIONAL DEVELOPMENT

IFLIP2 Fellow (Innovative Flipped Learning Instructional Project)	Aug. 2025 – Aug. 2028
• <u>Award ID</u>: NSF Award #2439762, IUSE program • Selected to participate in a three-year, multi-institutional STEM education reform cohort under the NSF-funded project “Developing STEM Education Reform Leaders through a Multidisciplinary and Cross-Institutional Community of Transformation”. Participating in cross-institutional workshops, symposia, and professional development activities with faculty from partner universities and community colleges.	

PEER-REVIEW ACTIVITIES

Journals: IEEE Transactions on Information Forensics and Security (TIFS, IF: 6.3), IEEE Transactions on Industrial Informatics (TII, IF: 12.3), IEEE Internet of Things Journal (IoT-J, IF: 10.2), IEEE Transactions on Big Data (TBD, IF: 7.5), IEEE Transactions on Wireless Communications (TWC, IF: 8.9), IEEE Transactions on Dependable and Secure Computing (TDSC, IF: 7.3), IEEE Transactions on Vehicular Technology (TVT, IF: 6.8), Elsevier Computers & Security (IF: 4.8).

Conferences: IEEE International Conference on Communications (ICC), Blockchain Research & Applications for Innovative Networks and Services (BRAINS), IEEE Wireless Communications and Networking Conference (WCNC), Applied Cryptography and Network Security (ACNS), International Conference on Network and Service Management (CNSM), IEEE SmartGridComm, IEEE International Conference on Metaverse Computing, Networking and Applications (MetaCom), IEEE International Conference on Digital Health (ICDH).

PROFESSIONAL CERTIFICATIONS

- Completed and obtained Microsoft: Security, Compliance, and Identity Fundamentals Certification (**SC-900 Credential**).
- Completed and obtained Microsoft: Azure Fundamentals Certification (**AZ-900 Credential**).
- Completed and obtained Microsoft: Azure Artificial Intelligence Fundamentals Certification (**AI-900 Credential**).

LEADERSHIP AND VOLUNTARY ACTIVITIES

Leadership Roles

- Elected **Secretary** of BSO (an official FIU student organization). Aug. 2023 - Jul. 2024
- Elected **Registered Student Organizations** of **FIU IEEE E-board**. Aug. 2022 - Jul. 2023
- Elected **Registered Student Organizations** of BSO. Aug. 2022 - Jul. 2023
- Chief Technical Coordinator of **IEEE KUET Student Branch**. June 2017 - Feb. 2018
- Former trainer of the software/web application development club **BitToByte**. June 2016 - Apr. 2017
- Former trainer at Bangladesh physics olympiad.

Conference Volunteering

- 2023 IEEE/IFIP Network Operations and Management Symposium (NOMS).
- 2017 3rd International Conference on Electrical Information and Communication Technology (EICT).